

Name: _____

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Wave Phenomena

Basic Properties

Types of Waves

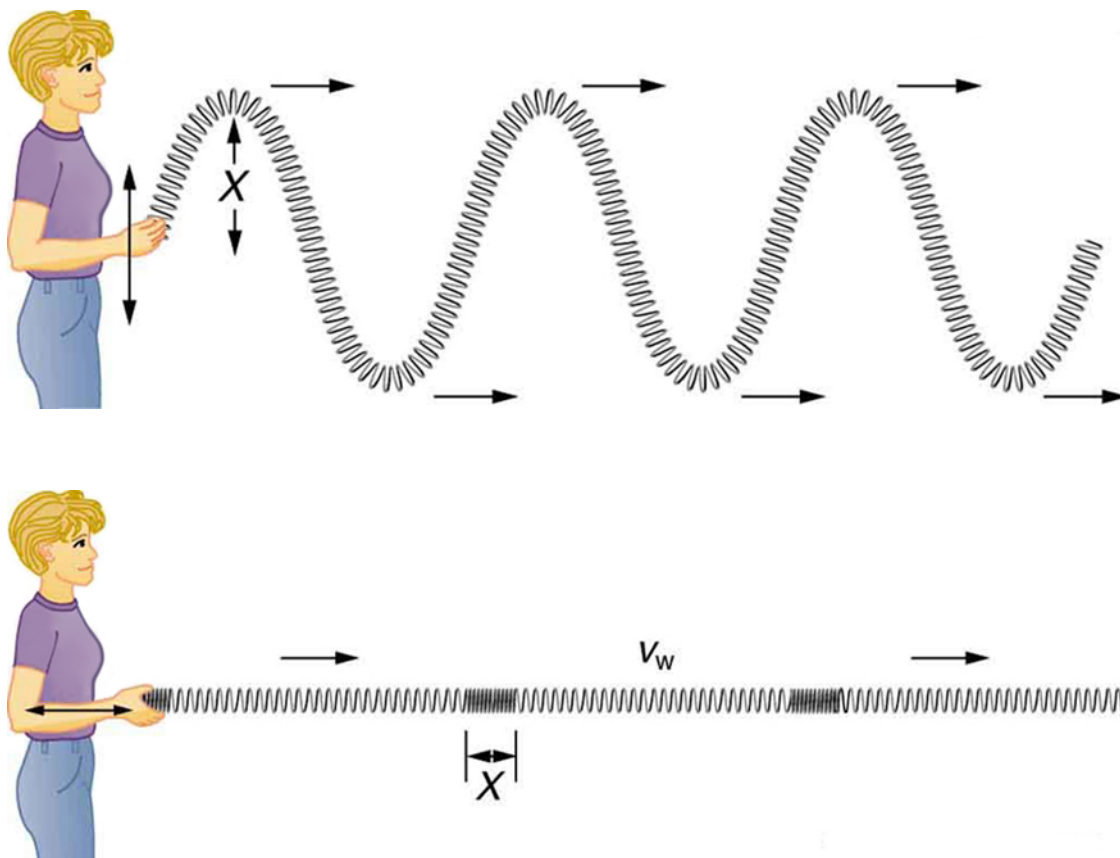


Image Credit: OpenStax *College Physics 2e*. Authored by: P.P Urone, R. Hinrichs, et al.

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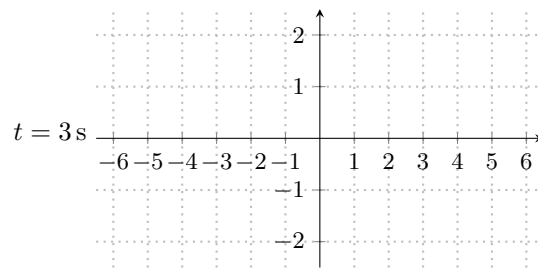
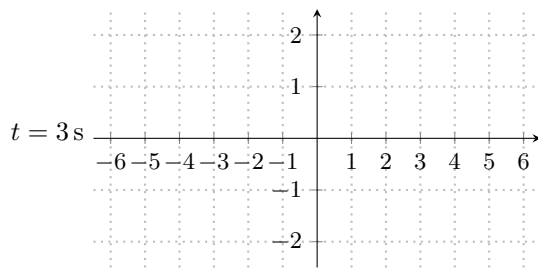
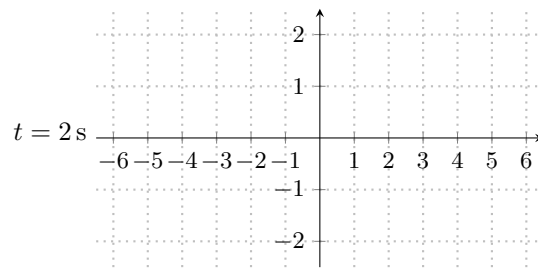
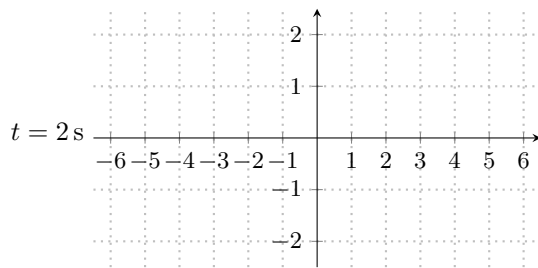
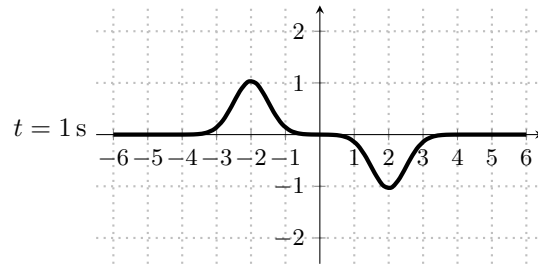
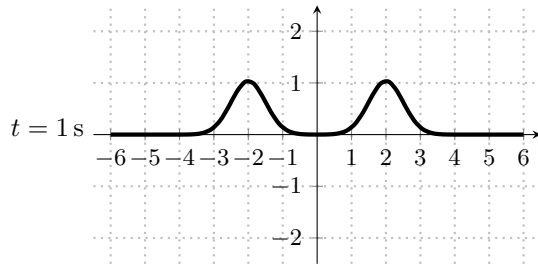
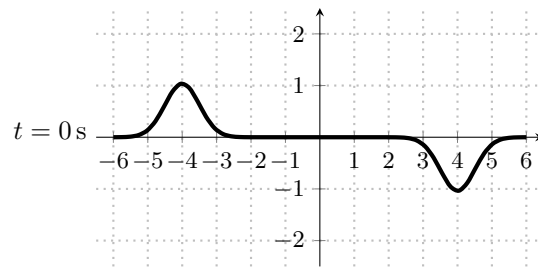
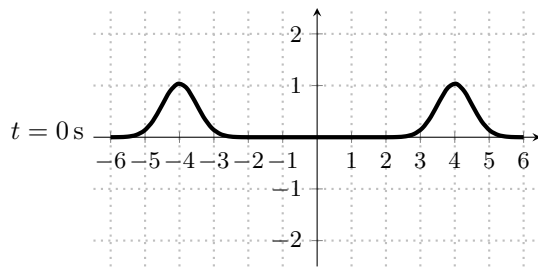
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Superposition



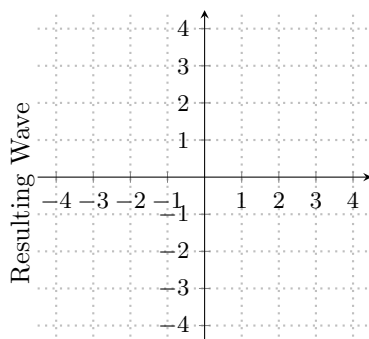
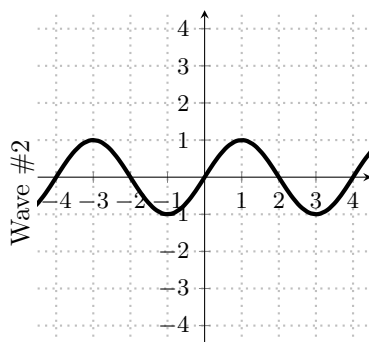
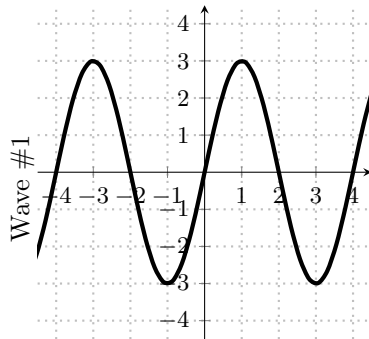
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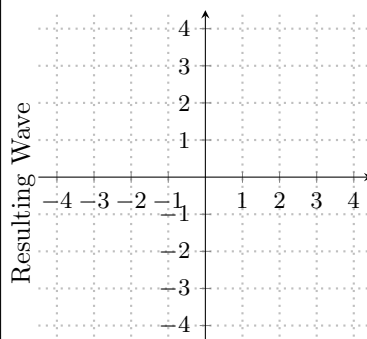
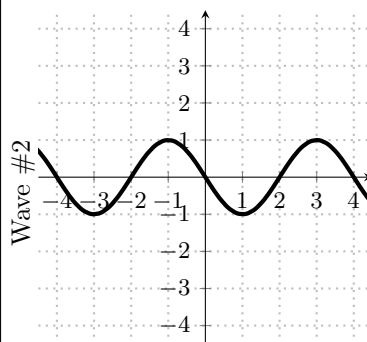
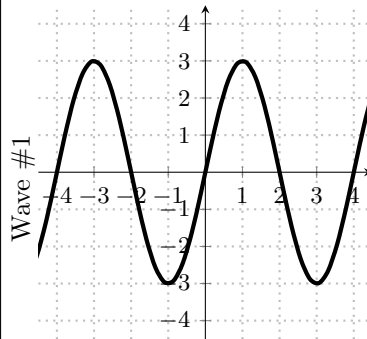
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Interference

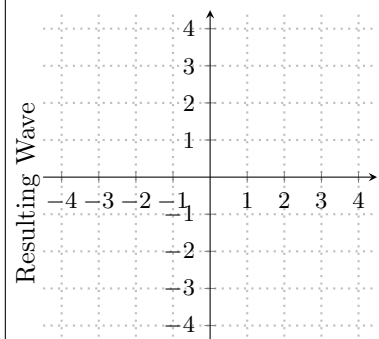
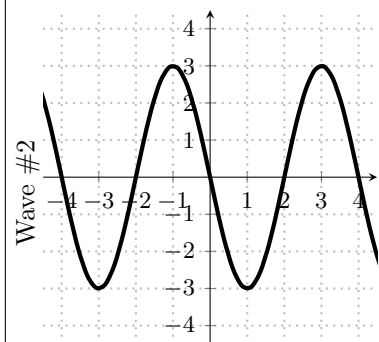
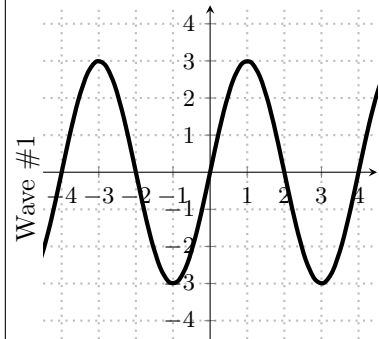
Situation (a)



Situation (b)



Situation (c)

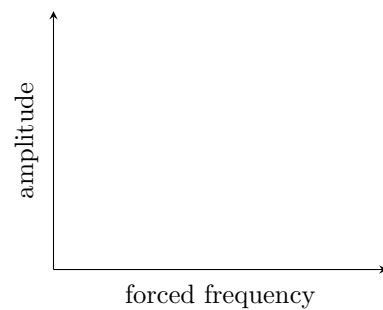
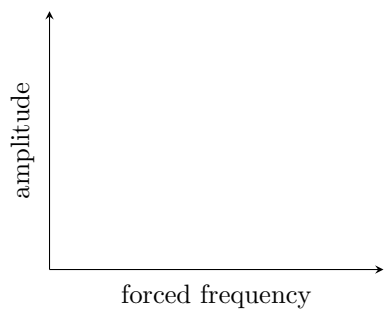


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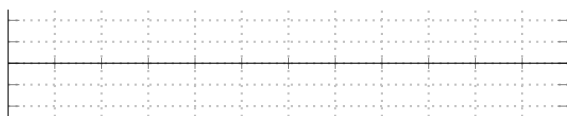
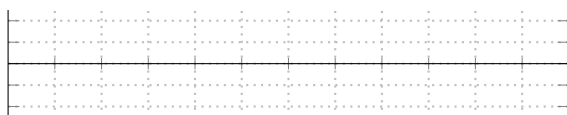
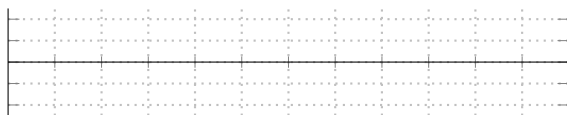
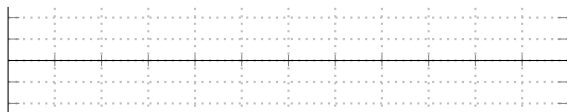
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Resonance



Standing Waves

“Normal Modes”; “Harmonics”; “Overtones”



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Example Problem The lowest note an open trumpet plays is C (concert B $\flat$) at 466.16 Hz. It turns out that this is actually the second harmonic of the air column in the trumpet

- Calculate the fundamental frequency of the trumpet.
- Calculate the first eight harmonics of the trumpet.
- Use the table below to determine which notes these harmonics correspond to.

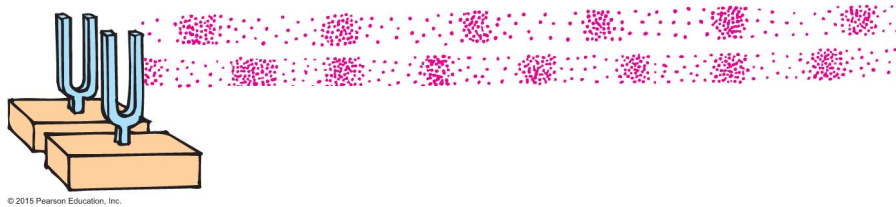
Concert Pitch	B $\flat$ Trumpet Pitch	Frequency (Hz)	Concert Pitch	B $\flat$ Trumpet Pitch	Frequency (Hz)	Concert Pitch	B $\flat$ Trumpet Pitch	Frequency (Hz)
B $\flat_3$	C $_4$	233.08	B $\flat_4$	C $_5$	466.16	B $\flat_5$	C $_6$	932.33
B $_3$	D $\flat_4$	246.94	B $_4$	D $\flat_5$	493.88	B $_5$	D $\flat_6$	987.77
C $_4$	D $_4$	261.63	C $_5$	D $_5$	523.25	C $_6$	D $_6$	1046.50
D $\flat_4$	E $\flat_4$	277.18	D $\flat_5$	E $\flat_5$	554.37	D $\flat_6$	E $\flat_6$	1108.73
D $_4$	E $_4$	293.66	D $_5$	E $_5$	587.33	D $_6$	E $_6$	1174.66
E $\flat_4$	F $_4$	311.13	E $\flat_5$	F $_5$	622.25	E $\flat_6$	F $_6$	1244.51
E $_4$	G $\flat_4$	329.63	E $_5$	G $\flat_5$	659.25	E $_6$	G $\flat_6$	1318.51
F $_4$	G $_4$	349.23	F $_5$	G $_5$	698.46	F $_6$	G $_6$	1396.91
G $\flat_4$	A $\flat_4$	369.99	G $\flat_5$	A $\flat_5$	739.99	G $\flat_6$	A $\flat_6$	1479.98
G $_4$	A $_4$	392.00	G $_5$	A $_5$	783.99	G $_6$	A $_6$	1567.98
A $\flat_4$	B $\flat_4$	415.30	A $\flat_5$	B $\flat_5$	830.61	A $\flat_6$	B $\flat_6$	1661.22
A $_4$	B $_4$	440.00	A $_5$	B $_5$	880.00	A $_6$	B $_6$	1760.00
						B $\flat_6$	C $_7$	1864.66

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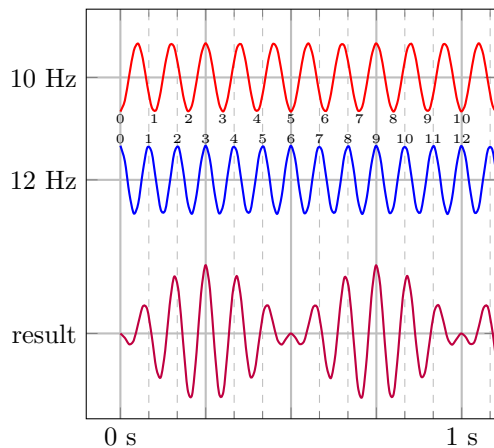
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Beats



Take a look at the illustration below that refers to a wave of frequency 10 Hz being played at the same time as a wave of frequency 12 Hz.



Draw your own beats! Draw two bugs jumping on the water. One bug jumps forward 3 cm each hop; the other bug jumps forward 4 cm each hop.

